Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: *To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration*	**Mission:** *To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky.
12/03/26

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Session	**2025-26 (EVEN)**	**Course Name**	**JAVA FSD Lab**
Semester	**6**	**Course Code**	**23IOT1607**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**04**
Course Outcome	1. Apply OOP concepts to solve the real-life problem. 2. Use the database concepts to integrate in giving OOP oriented solution. 3. Develop full-stack applications integrating frontend, backend, and database with security. 4. Design and develop mini-project with frontend, backend, and database integration, including authentication and testing.
Aim	To perform database Insert, Update, and Delete operations using Prepared Statement in Java with JDBC.

Problem Definition	1) To Insert the record
	
	Prepared Statement stmt=con.prepareStatement("insert into Emp
	values(?,?)");
	stmt.setInt(1,101);//1 specifies the first parameter in the query
	stmt.setString(2,"Ratan");
	
	int i=stmt.executeUpdate();
	System.out.println(i+" records inserted");
	
	2) To Update
	Prepared Statement stmt=con.prepareStatement("update emp set
	name=? where id=?");
	stmt.setString(1,"Sonoo");//1 specifies the first parameter in the
	query i.e. name
	stmt.setInt(2,101);

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	int i=stmt.executeUpdate();
	System.out.println(i+" records updated");
	3) To Delete
	
	Prepared Statement stmt=con.prepareStatement("delete from emp
	where id=?");
	stmt.setInt(1,101);
	int i=stmt.executeUpdate();
	System.out.println(i+" records deleted");

Theory (100 words)	Prepared Statement in Java JDBC is used to perform database Insert, Update, and Delete operations in a secure and efficient way. It allows SQL queries to be written with placeholders (?), and values are assigned later using methods like setInt() and setString(). This improves performance because the query is precompiled and can be reused multiple times. It also helps prevent SQL injection attacks, making database operations safer than normal Statement. The executeUpdate() method is used to execute these queries and returns the number of affected rows. Thus, PreparedStatement is widely used for reliable, faster, and secure database record manipulation in Java applications.
Procedure and Execution (100 Words)	First, import the required JDBC packages and establish a connection with the database using DriverManager. Create a PreparedStatement object by passing the SQL query with placeholders. For inserting records, set values using setter methods and execute the query using executeUpdate(). For updating records, prepare an update query, set new values and conditions, then execute it. For deleting records, prepare a delete query with a condition, set the required parameter, and execute. The executeUpdate() method returns the number of affected rows. Finally, display the result and close the PreparedStatement and database connection to release resources properly.

Code:

1. DBConnection.java

```
import java.sql.Connection;
import java.sql.DriverManager;

public class DBConnection {

    public static Connection getConnection() {

        Connection con = null;

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/studentdb",
                    "root",
                    "7421"
            );

        } catch (Exception e) {
            System.out.println(e);
        }

        return con;
    }
}
```

2. Main.java

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        StudentDAO dao = new StudentDAO();

        while (true) {
```

```
System.out.println("\n----- Student Registration System -----");
System.out.println("1. Register Student");
System.out.println("2. View Students");
System.out.println("3. Update Student");
System.out.println("4. Delete Student");
System.out.println("5. Exit");

System.out.print("Enter Choice: ");
int choice = sc.nextInt();

switch (choice) {

  case 1:

    System.out.print("Enter ID: ");
    int id = sc.nextInt();

    sc.nextLine();

    System.out.print("Enter Name: ");
    String name = sc.nextLine();

    System.out.print("Enter Marks: ");
    int marks = sc.nextInt();

    Student s = new Student(id, name, marks);

    dao.insertStudent(s);

    break;

  case 2:

    dao.displayStudents();

    break;

  case 3:

    System.out.print("Enter ID to Update: ");
    int uid = sc.nextInt();
```

```
                System.out.print("Enter New Marks: ");
                int newMarks = sc.nextInt();

                dao.updateStudent(uid, newMarks);

                break;

            case 4:

                System.out.print("Enter ID to Delete: ");
                int did = sc.nextInt();

                dao.deleteStudent(did);

                break;

            case 5:

                System.out.println("Program Ended");
                System.exit(0);

            default:

                System.out.println("Invalid Choice");

            }
        }
    }
}
```

3. Student.java

```
public class Student {

    private int id;
    private String name;
    private int marks;

    public Student(int id, String name, int marks) {
        this.id = id;
        this.name = name;
        this.marks = marks;
```

```
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public int getMarks() {
        return marks;
    }
}
```

4. StudenDAO.java

```
import java.sql.*;

public class StudentDAO {

    Connection con;

    // Constructor
    public StudentDAO() {
        con = DBConnection.getConnection();

        if (con != null) {
            System.out.println("Database Connected Successfully!");
        } else {
            System.out.println("Database Connection Failed!");
        }
    }

    // Insert
    public void insertStudent(Student s) {

        if (con == null) {
            System.out.println("Connection is null. Database not
connected.");
            return;
        }

        try {
```

```
            String query = "INSERT INTO student VALUES(?,?,?)";

            PreparedStatement ps = con.prepareStatement(query);

            ps.setInt(1, s.getId());
            ps.setString(2, s.getName());
            ps.setInt(3, s.getMarks());

            ps.executeUpdate();

            System.out.println("Student Registered Successfully");

        } catch (Exception e) {
            System.out.println(e);
        }
    }

    // Display
    public void displayStudents() {

        if (con == null) {
            System.out.println("Connection is null. Database not
connected.");
            return;
        }

        try {

            String query = "SELECT * FROM student";

            Statement st = con.createStatement();

            ResultSet rs = st.executeQuery(query);

            while (rs.next()) {

                System.out.println(
                    rs.getInt("id") + " " +
                        rs.getString("name") + " " +
                        rs.getInt("marks")
                );
            }
```

```
        } catch (Exception e) {
            System.out.println(e);
        }
    }

    // Update
    public void updateStudent(int id, int marks) {

        if (con == null) {
            System.out.println("Connection is null. Database not
connected.");
            return;
        }

        try {

            String query = "UPDATE student SET marks=? WHERE
id=?";

            PreparedStatement ps = con.prepareStatement(query);

            ps.setInt(1, marks);
            ps.setInt(2, id);

            int rows = ps.executeUpdate();

            if (rows > 0) {
                System.out.println("Student Updated");
            } else {
                System.out.println("Student ID not found");
            }

        } catch (Exception e) {
            System.out.println(e);
        }
    }

    // Delete
    public void deleteStudent(int id) {

        if (con == null) {
            System.out.println("Connection is null. Database not
```

```
connected.");
            return;
        }

        try {

            String query = "DELETE FROM student WHERE id=?";

            PreparedStatement ps = con.prepareStatement(query);

            ps.setInt(1, id);

            int rows = ps.executeUpdate();

            if (rows > 0) {
                System.out.println("Student Deleted");
            } else {
                System.out.println("Student ID not found");
            }

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Output:

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```
Run   Main
"C:\Program Files\Java\jdk-23\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 202
Database Connected Successfully!

----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 1
Enter ID: 63
Enter Name: Sheikh wasimuddin
Enter Marks: 99
java.sql.SQLIntegrityConstraintViolationException: Duplicate entry '63' for key 'student.PRIMARY'

----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 1
Enter ID: 67
Enter Name: Tanmay
Enter Marks: 91
java.sql.SQLIntegrityConstraintViolationException: Duplicate entry '67' for key 'student.PRIMARY'
```

```
Run   Main
----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 1
Enter ID: 69
Enter Name: Vedant
Enter Marks: 88
Student Registered Successfully

----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 2
63 Sheikh Wasimuddin 99
67 Tanmay 91
69 Vedant 88
```

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```
----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 3
Enter ID to Update: 69
Enter New Marks: 95
Student Updated

----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 2
63 Sheikh Wasimuddin 99
67 Tanmay 91
69 Vedant 95
```

```
----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 4
Enter ID to Delete: 67
Student Deleted

----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 2
63 Sheikh Wasimuddin 99
69 Vedant 95

----- Student Registration System -----
1. Register Student
2. View Students
3. Update Student
4. Delete Student
5. Exit
Enter Choice: 5
Program Ended
```

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<table>
<tr>
<td></td>
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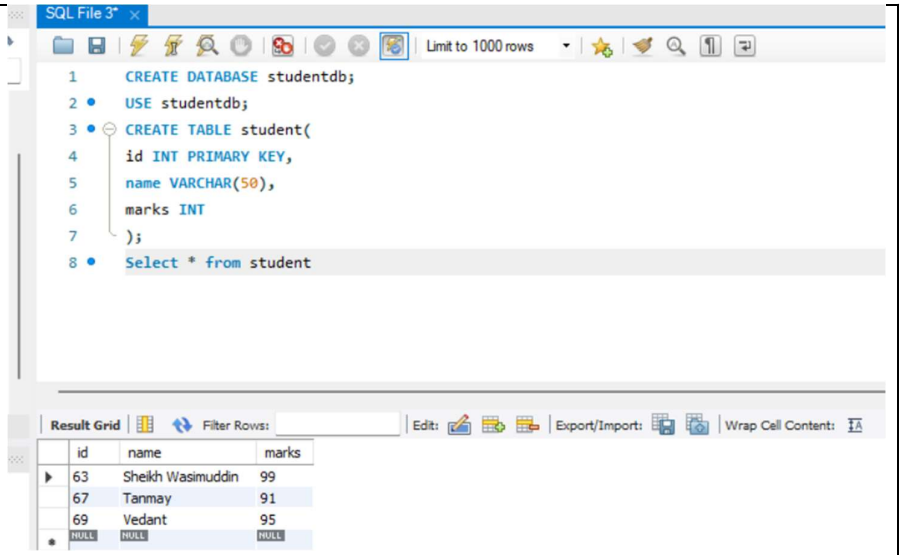

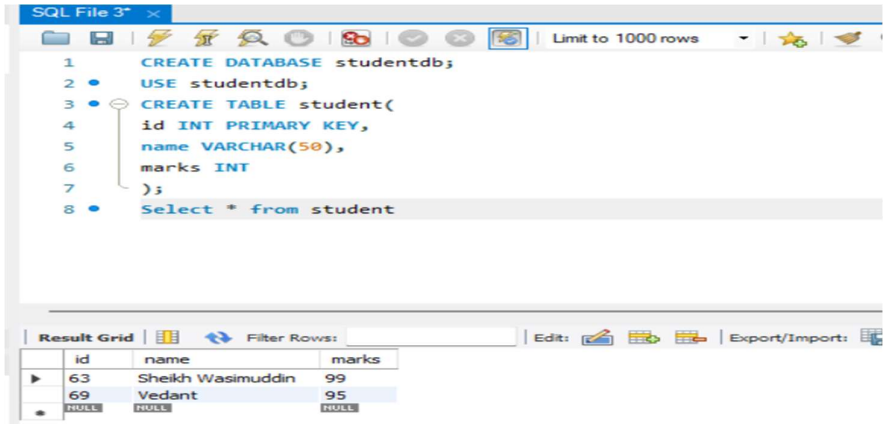

</td>
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<tr>
<td>Output Analysis</td>
<td>The program successfully performs Insert, Update, and Delete operations on the database using PreparedStatement in Java JDBC. During the Insert operation, a new record is added to the table and the output shows the number of records inserted. In the Update operation, the existing record is modified based on the given condition, and the output confirms how many records were updated. In the Delete operation, the selected record is removed from the table, and the output displays the number of records deleted. This proves that PreparedStatement executes SQL queries correctly, securely,</td>
</tr>
</table>

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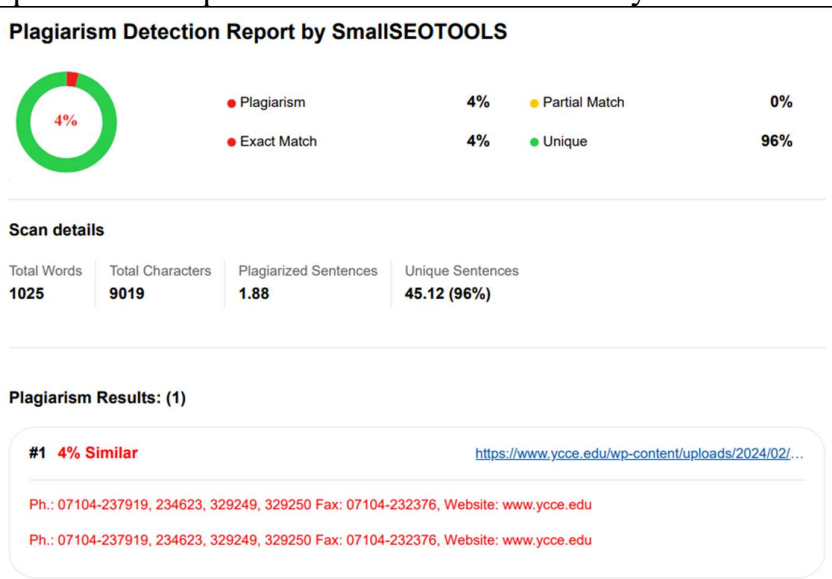
	and efficiently while providing confirmation of each database operation.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	From this program, we conclude that PreparedStatement in Java JDBC is an effective method for performing Insert, Update, and Delete operations on a database. It makes database handling more secure, efficient, and reliable by using placeholders and setting values dynamically. It also reduces the chances of SQL injection and improves execution speed for repeated queries. The successful output of inserted, updated, and deleted records shows that PreparedStatement is highly useful for real-time database applications and is an important concept in Java database connectivity.
Plag Report (Similarity index < 12%)	
Date	**12-03-26**

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